THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Formation of new substance indicates chemical change.
2	**(a)** Both A and R are true and R is the correct explanation of A.	H^+ is needed for acidic behaviour.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Kerosene prevents contact with air/water.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Weak forces need little energy.
5	**(a)** Both A and R are true and R is the correct explanation of A.	Low oxygen requires fast breathing.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Spinal cord reflex centre.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Copying errors produce small variations, which explains the statement.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Fusion with X or Y sperm decides the sex.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The properties of the convex mirror explain its use.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R does not explain why the image is inverted.
11	**(a)** Both A and R are true and R is the correct explanation of A.	High resistance draws almost no current.
12	**(a)** Both A and R are true and R is the correct explanation of A.	At a crossing there would be two directions of the field.
13	**(a)** Both A and R are true and R is the correct explanation of A.	The energy left is too little to support further levels.
14	**(d)** 44 g	Mass is conserved: 12 + 32 = 44 g.
15	**(b)** 7	Complete neutralisation to NaCl.
16	**(b)** Y	Reacting with cold water indicates more reactivity.
17	**(b)** $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$	Atoms of C, H and O balance.
18	**(c)** 6	$C_6H_{12}O_6 + 6\,O_2 \longrightarrow 6\,CO_2 + 6\,H_2O$.
19	**(d)** diabetes	Insulin.
20	**(c)** variation, and those from the cutting are identical to the parent	Sexual gives variation; vegetative does not.
21	**(c)** 50	1/4 of 200 = 50 (Rryy).
22	**(d)** 11.25 cm	1/f = -1/45 - 1/15 = -4/45, so f = -11.25 cm.
23	**(c)** -25 cm	f = -(far point) for objects at infinity.
24	**(a)** 6 ohm	R2 = rho 2L / 4A = R / 2.
25	**(d)** 90 degrees	Perpendicular is the maximum.
26	**(d)** 5000 J	5 x 10 x 10 x 10.
27	**(c)** copper	CuO is reduced to Cu by hydrogen.
28	**(d)** plaster of Paris	It sets into gypsum.
29	**(a)** CuO and CO_2	Calcination: $CuCO_3 \longrightarrow CuO + CO_2$.
30	**(b)** ethene	C_2H_4 = 28.
31	**(c)** uses carbon dioxide and releases oxygen	Photosynthesis replenishes oxygen.

32	**(a)** adrenaline	Fight or flight.
33	**(b)** undifferentiated cells to produce missing parts	Specialised dividing cells are essential.
34	**(d)** 1 : 1 : 1 : 1	Four gamete types x one.
35	**(a)** convex, f = 12 cm	Object at 2F.
36	**(b)** smaller wavelengths are scattered more by small particles	Rayleigh type scattering.
37	**(b)** V against I	Ohm's law.
38	**(b)** 100 Hz	Proportional to speed.
39	**(c)** UV radiation on oxygen	O2 splits and recombines.
40	**(a)** Both A and R are true and R is the correct explanation of A.	Energy is needed to break bonds.
41	**(a)** Both A and R are true and R is the correct explanation of A.	$2\,\mathrm{Mg} + \mathrm{O_2} \longrightarrow 2\,\mathrm{MgO}$: two reactants give one product, so R explains A.
42	**(a)** Both A and R are true and R is the correct explanation of A.	Without water the acid does not ionise, so no $\mathrm{H^+}$ is available to change the litmus; R explains A.
43	**(a)** Both A and R are true and R is the correct explanation of A.	Kerosene cuts off air and moisture, preventing the violent reaction.
44	**(a)** Both A and R are true and R is the correct explanation of A.	Self-linking into long chains, branches and rings (catenation) together with four bonds per atom gives a huge number of compounds.
45	**(a)** Both A and R are true and R is the correct explanation of A.	Turgid guard cells bulge apart and open the pore; shrunken ones close it.
46	**(a)** Both A and R are true and R is the correct explanation of A.	The short reflex arc through the spinal cord saves time, which makes the response quick.
47	**(a)** Both A and R are true and R is the correct explanation of A.	With a single parent and no gamete fusion there is no mixing of DNA, so the offspring are almost identical clones.
48	**(a)** Both A and R are true and R is the correct explanation of A.	Every F1 plant is Tt; the dominant allele T masks t, so R explains A.
49	**(a)** Both A and R are true and R is the correct explanation of A.	The wide field of view of the diminished, erect image lets the driver see more traffic behind.
50	**(a)** Both A and R are true and R is the correct explanation of A.	In myopia the image forms in front of the retina; the diverging lens shifts it back onto the retina.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the deflection is evidence of the field, not its explanation.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Decomposition recycles nutrients.
3	**(d)** 16 g	40 - 24 = 16 g.
4	**(b)** 25 mL	Ratio 1 : 2.
5	**(d)** S	Each displaces the next, so S is lowest.
6	**(a)** 6	Given in the equation.
7	**(b)** 1	$6\,CO_2 + 6\,H_2O \longrightarrow C_6H_{12}O_6 + 6\,O_2$.
8	**(d)** reflex action is faster than brain sensation	Information reaches the brain later.
9	**(b)** spores need moisture to germinate	Water is needed to activate spores.
10	**(a)** 1/4	Each birth is independent.
11	**(a)** -40 cm	m = -v/u = 2 gives v = +40; 1/f = 1/40 - 1/20 = -1/40.
12	**(b)** -1 D and +1.5 D	Distance: f = -100 cm. Reading: 1/f = -1/40 + 1/25 = 3/200, P = +1.5 D.
13	**(d)** 9 R	L is tripled, A is one third.
14	**(b)** zero	No change of linked field.
15	**(d)** 40 kJ	Two transfers.
16	**(d)** zinc	Zn is more reactive; $ZnSO_4$ is colourless.
17	**(a)** $CuSO_4 \cdot 5\,H_2O$	Blue vitriol.
18	**(c)** copper	$CuO + H_2 \longrightarrow Cu + H_2O$.
19	**(c)** ethanoic acid	Vinegar.
20	**(c)** chlorophyll is needed for photosynthesis	Chlorophyll.
21	**(b)** iodine	Goitre.
22	**(b)** shortage of nutrients and oxygen	The placenta is the exchange site.
23	**(a)** new combinations of independently inherited genes	Independent assortment gives new combinations.
24	**(c)** 1.5 f	1/v = 1/f - 1/3f = 2/3f.
25	**(b)** blue light more strongly than red light	Shorter wavelengths are scattered more.
26	**(b)** 5 ohm	R = V / I.
27	**(b)** 3 E	Proportional to rate of change.
28	**(d)** lower levels have more energy	Less loss in the chain.
29	**(c)** A is true but R is false.	A is true; R is false because nitrogen is unreactive.
30	**(a)** Both A and R are true and R is the correct explanation of A.	No free ions are produced.
31	**(a)** Both A and R are true and R is the correct explanation of A.	Layers repel and crystal breaks.
32	**(c)** A is true but R is false.	A true; in graphite each carbon bonds to three others, so R is false.
33	**(c)** A is true but R is false.	Bile has no enzymes; it emulsifies fats.
34	**(c)** A is true but R is false.	A true; R false, roots grow downward.
35	**(a)** Both A and R are true and R is the correct explanation of A.	The scrotum keeps the testes cooler.
36	**(a)** Both A and R are true and R is the correct explanation of A.	Masking explains absence of the recessive trait.

37	**(d)** A is false but R is true.	A is false (image is virtual); R is true.
38	**(a)** Both A and R are true and R is the correct explanation of A.	Higher n means more bending.
39	**(a)** Both A and R are true and R is the correct explanation of A.	Greater collisions give more resistance.
40	**(a)** Both A and R are true and R is the correct explanation of A.	Needed for a temporary magnet.
41	**(a)** Both A and R are true and R is the correct explanation of A.	A low-resistance meter in series does not change the current it is measuring.
42	**(a)** Both A and R are true and R is the correct explanation of A.	If two lines crossed, there would be two directions of the field at the point of intersection.
43	**(a)** Both A and R are true and R is the correct explanation of A.	The simple substances released are taken up again by the producers.
44	**(b)** Both A and R are true but R is not the correct explanation of A.	$\mathrm{BaCl_2 + Na_2SO_4 \longrightarrow BaSO_4 + 2\,NaCl}$; both true, but the white precipitate is due to the insolubility of $\mathrm{BaSO_4}$, not because the reaction is called double displacement.
45	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the disinfecting action is due to the chlorine it releases, not due to the way it is manufactured.
46	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true: $\mathrm{Al_2O_3}$ reacts with acids and bases; utensils use depends on good conductivity of heat and malleability, so R does not explain A.
47	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R is an additional fact about another oxidising agent and does not explain A.
48	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; arteries carry blood under high pressure (hence thick walls) while valves are a separate feature of veins.
49	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; they are different hormones, so R does not explain A.
50	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; they are different modes of asexual reproduction, and R does not explain A.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** the ciliary muscles changing the shape of the lens	Accommodation by the ciliary muscles.
2	**(c)** R / n^2	Each part R/n; parallel gives R/n^2.
3	**(a)** develops an induced current while the magnet moves	A changing field links the coil only during the motion.
4	**(a)** insects and eagles	Insects (T2) and eagle (T5).
5	**(a)** Na_2CO_3	Carbonates release CO_2.
6	**(c)** $NaHCO_3$	Baking soda.
7	**(b)** copper	Cu and CuO.
8	**(a)** ethanol	Dehydration and oxidation.
9	**(a)** both plants	No light or no CO2 means no photosynthesis.
10	**(c)** gigantism	Excess growth hormone gives excessive growth.
11	**(b)** The egg cannot meet the sperm	Fertilisation occurs in the oviduct.
12	**(b)** environment affects expression of the trait	Phenotype depends on environment.
13	**(d)** +10 cm	Convex magnifier: m = 4, v = -30, u = -7.5; 1/f = -1/30 + 1/7.5 = 1/10.
14	**(b)** larger than the wavelength of light	All colours scatter.
15	**(c)** 1.5 A	R = 2 + 2 = 4 ohm.
16	**(a)** zero	Positive and negative halves cancel.
17	**(c)** Biodegradable waste	It is easily decomposed.
18	**(a)** Both A and R are true and R is the correct explanation of A.	Equal atoms mean equal mass.
19	**(a)** Both A and R are true and R is the correct explanation of A.	Mild base neutralises acid.
20	**(a)** Both A and R are true and R is the correct explanation of A.	Delocalised electrons carry the current.
21	**(a)** Both A and R are true and R is the correct explanation of A.	Scum forms.
22	**(a)** Both A and R are true and R is the correct explanation of A.	Transpiration pull.
23	**(a)** Both A and R are true and R is the correct explanation of A.	Prevents goitre.
24	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A.
25	**(a)** Both A and R are true and R is the correct explanation of A.	Equal gametes T and t give TT : Tt : tt = 1 : 2 : 1.
26	**(a)** Both A and R are true and R is the correct explanation of A.	A thin parallel-sided region gives no net deviation.
27	**(a)** Both A and R are true and R is the correct explanation of A.	Shorter wavelength scatters more.
28	**(a)** Both A and R are true and R is the correct explanation of A.	Parallel arrangement features.
29	**(d)** A is false but R is true.	A is false: the force is zero when the conductor is parallel to the field; R is true.
30	**(a)** Both A and R are true and R is the correct explanation of A.	Biomagnification.
31	**(d)** 40 mL	Ratio H2 : O2 = 2 : 1.

32	**(b)** The first is more acidic	Lower pH means a more acidic solution.
33	**(c)** $\mathrm{Na_2O}$	Soluble basic oxides form alkalis.
34	**(c)** carbon dioxide and water	Complete combustion of a carbon compound gives CO2 and H2O.
35	**(a)** 5040	70 x 72.
36	**(c)** not feel a stimulus	Information does not reach CNS.
37	**(d)** 16	2^4 = 16.
38	**(d)** 1/4	Four equally likely types.
39	**(a)** 2	n = 3/1.5.
40	**(a)** white light	The second prism reverses the dispersion.
41	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the second statement concerns the sex chromosome carried by sperm and does not explain the equal contribution.
42	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the ray falls normally on the mirror and retraces its path, which is not explained by $R = 2f$.
43	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; R describes how the rainbow forms but does not explain the required geometry of Sun, observer and rain.
44	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true ($R = \rho l/A$ and ρ changes with temperature), but R does not explain A.
45	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the field pattern outside is like that of a bar magnet, which is not explained by the uniform field inside.
46	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; R states the function of ozone, which is not the reason CFCs damage it.
47	**(c)** A is true but R is false.	Iron gains oxygen (oxidation), so A is true; iron is oxidised, not reduced, so R is false.
48	**(c)** A is true but R is false.	Enamel (calcium phosphate) is corroded below pH 5.5 by acids, formed when bacteria degrade sugar; R is false.
49	**(c)** A is true but R is false.	Galvanisation is true, but zinc is *more* reactive than iron and protects it even if the coating is broken; R is false.
50	**(c)** A is true but R is false.	Hydrogen adds across the double bonds, converting them into single bonds; so R is false.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.